



Business proposal

Pitch and bank-loan proposal for MetriQ — problem, product, market, prices, costs, timeline and funding ask.

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Summary

MTRobotics builds MetriQ: a camera inspection station that sits on an existing production line and removes bad products automatically. It finds new defect types without weeks of photo collection and model training (zero-shot detection), and it is installed in hours, without stopping the line.

We sell to small and mid-size food, agriculture and light manufacturing plants in Vietnam and Canada — plants that still check products by eye and cannot afford a USD 50,000+ vision system.

We ask for a **USD 120,000 loan** to build stock for 10 units, two demo stations, and the first paid installations. Each MetriQ sold at the proposed price returns about **USD 8,900 gross profit**; selling 4 units a year covers the loan repayment.

Item	Value
Product	MetriQ AI inspection station (live)
Markets	Vietnam (first), Canada
Proposed price	USD 9,000 (Lite) – USD 14,000 (Pro)
Hardware cost per unit	USD 2,133 (Lite) – USD 3,619 (Pro)
Gross margin per unit	About 63%
Customer payback	Under 12 months (Vietnam, 2 shifts)
Funding ask	USD 120,000, 5-year term
Monthly repayment (est.)	About USD 2,550 at 10% per year

The problem

- Most small and mid-size plants still inspect products by eye. People get tired, miss defects and judge differently from shift to shift.
- Existing vision systems are built for large factories. A mid-range system costs USD 50,000–150,000 plus USD 5,000–15,000 to install, and needs weeks of photo collection and tuning for each product.
- When the product or defect changes, the system has to be retrained. Small plants cannot wait or pay for that.

What we do

MetriQ is a complete inspection station: camera, lighting, edge computer, software and reject output, delivered and installed by one team.

1. Mount the station over the existing conveyor. No line changes, no downtime.
2. Tell MetriQ what a defect looks like. It starts finding it the same day.

3. MetriQ checks every product in real time and signals the reject system to remove bad ones.
4. The plant gets counts, defect types and reports for each shift.

What it does on the line:

- Finds defects and unusual products.
- Detects, tracks and counts products.
- Sorts and rejects bad products automatically.
- Produces shift and batch quality reports.

Published performance: 100+ parts per second, under 1 mm accuracy, more than 99.5% of defects caught.

Why we are different

What matters to the plant	Typical vision system	MetriQ
New defect type	Collect and label photos, retrain for weeks	Zero-shot detection — describe the defect, works the same day
Installation	Weeks, often a line stop	Hours, no downtime
Who you deal with	Camera brand + integrator	One team: hardware, software, install
Price	USD 50,000–150,000 + install	USD 9,000–14,000, installed
Natural products (fruit, nuts, seafood)	Hard — every item looks different	Built for variable products

Zero-shot detection means the AI can find a defect it was never trained on. The plant does not need to collect thousands of example photos first. This is what makes fast installation and low price possible.

Market

Market fact	Number
AI visual inspection, world	USD 29.8B (2025) → USD 85.2B (2030), 23% a year; Asia-Pacific grows fastest
Machine vision, Asia-Pacific	USD 6.3B (2025) → USD 9.8B (2030), 9% a year
Vietnam GDP growth 2025	8.0%; manufacturing grew 10.0%
Vietnam food processing	USD 88B in 2025, up 11%
Vietnam agriculture, forestry and fishery exports	USD 70.1B in 2025, up 12%
Canada food and beverage processing	USD 173.4B (2024), largest manufacturing sector, 8,800 companies
Vietnam manufacturing wage	About VND 8.4M (USD 332) per month (Q1 2025)

Why now:

- Vietnam's food and farm exports are growing fast, and export buyers demand consistent quality.
- Labour costs are rising and plants are looking for automation they can afford.
- AI that works without large training data has only become practical in the last two years. Large competitors still sell the old, slow-to-train approach.

Target customers

Segment	Country	Why they buy
Fruit, cashew, coffee, pepper processors	Vietnam	Export quality rules, many manual sorters
Seafood and food packing	Vietnam	Food safety, labour cost
Mid-size food and beverage plants	Canada	High labour cost, hard to hire inspectors
Light manufacturing (plastic, metal parts)	Vietnam, Canada	Surface defects checked by hand today

Sales channels: direct sales to plant owners, plus partners — consultancies that deploy camera AI (OCD) and machine builders that ship complete processing lines (PSL Machinery).

Products and prices

Product	What it is	Status	Proposed price (USD)
MetriQ Pro	GPU edge computer, zero-shot detection, full speed	Live	14,000 one-time, installed
MetriQ Lite	CPU computer, for simpler or slower lines	Live	9,000 one-time, installed
Support plan	Remote support, updates, new defect setup	Planned	1,200–1,500 per year
Paid pilot	4-week trial on the customer's line	Planned	Credited to the purchase
MTR-M	Autonomous mobile robot for moving materials	In development	Not priced yet
Robot arm	Pick-and-place arm	Work in progress	Not for sale

Market reference: entry-level AI inspection USD 3,000–10,000 (software or camera only), mid-range USD 50,000–150,000 plus installation. MetriQ is a complete installed station priced near the entry level.

Cost and margin

Hardware cost comes from the priced AI_BOX BOM (single unit, before shipping, duty and VAT).

Cost item (USD)	MetriQ Pro	MetriQ Lite
Edge computer + storage	2,075	589
Camera, lighting, cable	584	584

Cost item (USD)	MetriQ Pro	MetriQ Lite
Trigger sensor and reject I/O	230	230
Enclosure and mounting	300	300
Power and networking	125	125
Stack light, touchscreen, UPS	305	305
Hardware total	3,619	2,133
Shipping, duty, assembly (20%)	724	427
Installation and travel	800	800
Delivered cost	5,143	3,360
Proposed price	14,000	9,000
Gross profit	8,857	5,640
Gross margin	63%	63%

Customer payback (Vietnam): one station replaces about 2 inspectors per shift. Two shifts × 2 inspectors × USD 332 × 12 months = USD 15,936 saved per year. A Pro unit pays back in under 11 months. Payback is faster in Canada, where wages are higher.

Competition

Competitor	What they sell	Our edge
Cognex, Keyence (via integrators)	Smart cameras, needs integrator	One team, installed in hours, no training data
TOMRA, Satake, Chinese color sorters	Full optical sorting lines	Retrofit on existing lines at a fraction of the price; catches shape and surface defects color sorters miss
Musashi AI (Canada)	Turnkey AI for automotive parts	We serve food, agriculture and light manufacturing
FaceNet, SmartSys (Vietnam)	AI software, often bundled with MES	Complete station with hardware and reject output
UnitX, iFactory	AI inspection for large factories	Priced and supported for mid-size local plants

Timeline

When	Milestone
Q4 2026	Build 2 demo stations. Start 2 paid pilots with Vietnam food or farm processors.

When	Milestone
Q1 2027	First sales. Reject kit and shift reports ready. Support plan launched.
Q2 2027	First Canada pilot (Ontario food plant).
H2 2027	12 units installed in total. MTR-M mobile robot pilot.
2028	Partner sales through consultancies and machine builders. 30 units per year.

Funding request

We ask for **USD 120,000**, repaid over 5 years.

Use of funds	USD	Share
Stock for 10 stations (hardware, landed)	43,000	35.8%
2 demo stations (Vietnam, Canada)	12,000	10.0%
Engineering: reject kit, reports, installer	30,000	25.0%
Sales and marketing: trade shows, demos, travel	12,000	10.0%
Electrical safety certification (CE / CSA)	8,000	6.7%
Contingency	15,000	12.5%
Total	120,000	100%

Repayment

- Estimated payment: about USD 2,550 per month (5 years, 10% per year; rate to be set by the lender).
- Each Pro unit returns about USD 8,857 gross profit. **4 Pro units a year cover all loan payments.**
- Year-1 target is 12 units (8 Pro, 4 Lite): about USD 148,000 revenue and USD 93,000 gross profit.
- Stock bought with the loan is converted to cash as each unit is sold; paid pilots bring cash in before full sales.

Risks

Risk	How we handle it
Slow first sales	Paid pilots credited to purchase; demo stations at trade shows
Accuracy on a new product type	4-week pilot on the customer's own line before purchase
Hardware price or supply changes	Two computer options (Pro, Lite); common industrial brands
Small team	Standard installer and checklist; partners for sales and service
Currency (VND, CAD, USD)	Price and buy hardware in USD

Team

- **Thong Huynh — Founder.** Robotics engineer, B.Eng. Mechatronics (Ontario Tech University). Robotics Software Engineer at ABI Ltd. Research assistant at MARS Lab; design lead, Ontario Tech RoboMaster team.
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